

Retail Sales Analysis

Prepared By: Virang Raje

Tools Used:

- Python (Pandas, Matplotlib)
- PostgreSQL
- Power BI
- Excel

Dataset:

Online Retail II Dataset (UCI Machine Learning Repository)

1. Executive Summary

This project analyzes the Online Retail II dataset containing over one million retail transactions. The objective is to identify sales trends, customer purchasing behavior, top-performing products, and country-wise revenue distribution. The project uses Python for data cleaning and exploratory analysis, PostgreSQL for business analysis, and Power BI for dashboard creation.

The analysis provides insights that can help businesses improve inventory management, customer retention, and sales strategies.

2. Business Problem

Retail businesses generate large amounts of transaction data every day. Without proper analysis, it becomes difficult to answer important business questions such as:

- Which products generate the highest revenue?
- Who are the most valuable customers?
- Which countries contribute the most sales?
- During which months does sales performance improve?
- At what time of day do customers place the most orders?

This project addresses these questions using data analytics.

3. Objectives

The primary objectives of this project are:

- Clean and prepare raw retail transaction data.
- Remove invalid and duplicate records.
- Generate business KPIs.
- Analyze customer purchasing behavior.
- Identify top-performing products.
- Analyze monthly and yearly sales trends.
- Build an interactive Power BI dashboard.
- Generate actionable business recommendations.

4. Dataset Overview

Dataset Name: Online Retail II

Source: UCI Machine Learning Repository

Original Dataset

Metric	Value
Total Rows	1,067,371
Total Columns	8

Original Columns

- Invoice
- StockCode
- Description
- Quantity
- InvoiceDate
- Price
- Customer ID
- Country

5. Data Cleaning

The following cleaning steps were performed using Python:

- Removed duplicate records.
- Removed cancelled orders.
- Removed records with invalid quantity values.

- Removed records with invalid prices.
- Removed rows with missing product descriptions.
- Retained missing Customer IDs because anonymous purchases may still contain valuable sales information.
- Created additional analysis columns:
 - Total Sales
 - Year
 - Month
 - Month Number
 - Quarter
 - Day
 - Weekday
 - Hour

Final Dataset

Metric	Value
Final Rows	1,007,913
Final Columns	16

6. Exploratory Data Analysis (EDA)

The following KPIs were calculated:

KPI	Value
Total Revenue	\$20,476,260.45
Total Orders	40,079
Total Customers	5,878
Total Products	5,399
Average Order Value	\$510.90
Average Quantity Sold	11.12

Additional analyses included:

- Monthly Sales Trend
- Top Products
- Top Customers

- Country-wise Revenue
- Weekday Sales
- Hourly Sales

7. SQL Analysis

PostgreSQL was used to answer business questions such as:

- Total Revenue
- Total Orders
- Top Customers
- Top Products
- Monthly Revenue
- Revenue by Country
- Customer Lifetime Value (CLV)
- Window Functions
- Ranking
- Running Totals
- Month-over-Month Growth

These SQL queries helped identify important sales patterns and customer behavior.

8. Power BI Dashboard

An interactive Power BI dashboard was created with the following components:

- KPI Cards
- Monthly Sales Trend
- Top Products
- Top Customers
- Revenue by Country
- Revenue by Weekday
- Revenue by Hour
- Quarter-wise Revenue
- Interactive Slicers (Year, Quarter, Month, Country)

9. Key Findings

The analysis revealed several important insights:

- The United Kingdom generated the highest overall revenue.

- A relatively small group of customers contributed a significant share of total sales.
- Revenue increased during the holiday shopping season.
- Certain products consistently generated higher sales than others.
- Customer purchasing activity peaked during standard business hours.

10. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Maintain sufficient inventory for top-selling products to avoid stock shortages.
- Focus customer retention efforts on high-value customers through loyalty programs.
- Increase marketing activities during lower-performing months.
- Expand promotional campaigns in countries showing strong growth potential.
- Monitor hourly sales patterns to optimize staffing and warehouse operations.

12. Conclusion

This project demonstrates an end-to-end data analytics workflow, beginning with raw transactional data and ending with business insights and interactive visualizations. By combining Python, PostgreSQL, Excel, and Power BI, the project highlights practical skills in data cleaning, SQL analysis, dashboard development, and business reporting.

The findings can support data-driven decisions related to inventory planning, customer management, and sales strategy.